

AI ASSISTED CODING

Lab 4: Advanced Prompt Engineering – Zero-shot, One-shot, and Few-shot Techniques

NAME : G.Vijaya Simha Reddy

ROLLNO:2503A52671

BATCH:23

Python Code

Replace "YOUR_GEMINI_API_KEY" with your API key.

```
import google.generativeai as genai

# Configure API
genai.configure(api_key="YOUR_GEMINI_API_KEY")

model = genai.GenerativeModel("gemini-1.5-flash")

# Sample emails
emails = [
    "I was charged twice for my subscription this month.",
    "My app crashes every time I open it.",
    "I love the new update. Great work!",
    "Can you tell me your office timings?",
    "I need a refund for my last payment."
]

# ----- ZERO-SHOT -----
print("===== ZERO-SHOT =====")
```

```
zero_prompt = f"""
```

Classify the customer email into one of these categories:

Billing, Technical Support, Feedback, Others.

```
Email: "{emails[0]}"
```

```
Return only the category.
```

```
"""
```

```
response = model.generate_content(zero_prompt)
```

```
print("Email:", emails[0])
```

```
print("Category:", response.text.strip())
```

```
# ----- ONE-SHOT -----
```

```
print("\n==== ONE-SHOT =====")
```

```
one_prompt = f"""
```

Example:

Email: "My internet bill was charged twice."

Category: Billing

Now classify this email:

```
Email: "{emails[1]}"
```

```
Return only the category.
```

```
"""
```

```
response = model.generate_content(one_prompt)
```

```
print("Email:", emails[1])
```

```
print("Category:", response.text.strip())
```

```
# ----- FEW-SHOT -----
```

```
print("\n==== FEW-SHOT ====")
```

```
few_prompt = f"""
```

Examples:

Email: "My payment failed but money was deducted."

Category: Billing

Email: "The application crashes when I click Login."

Category: Technical Support

Email: "Your customer service was excellent."

Category: Feedback

Now classify this email:

Email: "{emails[4]}"

Return only the category.

```
"""
```

```
response = model.generate_content(few_prompt)
```

```
print("Email:", emails[4])
```

```
print("Category:", response.text.strip())
```

Expected Output

Email

Expected
Category

I was charged twice for my subscription this

Billing

month.

My app crashes every time I open it.

I love the new update. Great work!

Can you tell me your office timings?

I need a refund for my last payment.

Technical Support

Feedback

Others

Billing